Electrician Apprenticeship 2nd Year

Alessandro Volta (1745 - 1827)

Volta was the developer of the voltaic pile, the first electric battery. By determining zinc and silver were the most effective pair to produce electricity, he found the first method for generation of a sustained electrical current. He also invented the electrophorus (a device that produced a static electric charge), an electric condenser and the voltaic cell, plus he discovered methane. Napoleon made him a count. The Volt was named after him, and his likeness was placed on the now-obsolete 10000 Italian Lire banknote.

CONTENTS: August 2025 **10351-40**

THEORY

	Module		Title
1	030201a	02	Fundamentals of Alternating Current
2	030201b	02	Properties of Inductors and Capacitors
3	030201c	02	Inductors and Capacitors in Circuits
4	030201d	02	Principles of AC Circuits
5	030201e	02	Power Relationships
6	030202a	02	Series AC Circuits
7	030202b	02	Series RLC Circuits
8	030202c	02	Parallel ac Circuits
9	030202d	02	Parallel RLC Circuits
10	030202e	02	Single Phase Motors
11	030202f	02	Single Phase Power Factor Correction
13	10351-40	Aug 2025	Mod 1_AC Properties Workbook 1
14	10351-40	Aug 2025	Mod 2_AC Properties Workbook 2
15	10351-40	Aug 2025	Mod 3_Series RLC Workbook
16	10351-40	Aug 2025	Mod 4_Parallel RLC, PFC and Single Phase Motors Workbook
17	10351-40	Aug 2025	Mod 14_AC Fundamentals Lab
18	10351-40	Aug 2025	Mod 15_Series RLC Circuits Lab
19	10351-40	Aug 2025	Mod 16_Parallel RLC Circuits Lab
20	10351-40	Aug 2025	Mod 17_Single Phase Motors Lab

CODE

	Module		Title
32	030205a	02	Electric Vehicle Charging
33	030205b	02	Electric Space Heating and Cooling
34	030205c	02	Service Ampacity for Apartments and Similar Buildings
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36	030205e	02	Capacitor Bank Installations
37	030205f	02	Pools, Mobile Homes and Temporary Wiring
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40	030205i	02	Specifications
41	030205j	02	Drawings and Plans
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Electrical Systems

CONTROLS & SWITCHING (CSC)

	Module		Title
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22	030204b	02	Timers and Smart Relays
23	303204c	02	Pilot and Overcurrent Devices
24	030204d	02	Drawing and Diagram Conversion
25	030204e	02	Motor Starters
26	030204f	02	Single Motor Control
27	030204g	02	Reversing Motor Starters
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29	10351-40	Aug 2025	Mod 10_Timers and Smart Relays
30	10351-40	Aug 2025	Mod 11_Magnetic Motor Starters
31	10351-40	Aug 2025	Mod 12_Reversing Magnetic Motor Starters

HEATING AND COOLING

	Module		Title
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45	030203b	02	Forced-Air Heating Systems
46	030203c	02	Cooling Systems
47	030203d	02	HVAC Rooftop Units
48	030203e	02	Basic Hot Water Heating Systems
49	030203f	02	Heat Trace
52	10351-40	Aug 2025	Mod 05_Forced Air Furnace LAB
53	10351-40	Aug 2025	Mod 06_High-Efficient Furnaces LAB
54	10351-40	Aug 2025	Mod 07_Cooling Systems LAB
55	10351-40	Aug 2025	Mod 08_Air Handling Unit LAB
56	10351-40	Aug 2025	Mod 09_Hot Water Heating Systems LAB